

Inorganic Chemistry Practical 3: Qualitative Analysis — Anion Identification & Gas Testing

Halide Discrimination, Oxoanion Confirmation, and Analytical Gas Detection

CANDIDATE NAME	URWAH	CENTRE NUMBER	PK082
COMPONENT	Paper 3 — Advanced Practical Skills (Inorganic Chemistry Suite)	CANDIDATE NUMBER	0142
TIME ALLOWED	2 Hours	TOTAL MARKS	40 Marks

INSTRUCTIONS TO CANDIDATES:

- Answer all questions in the spaces provided on the question paper.
- Give working for all calculations and record all experimental readings to the required precision:
 - Burette readings to the nearest 0.05 cm³; balance readings to 0.01 g / 0.001 g.
 - Thermometer readings to the nearest 0.5 °C.
- You may use an HB pencil for any rough working, graphs, or apparatus diagrams.
- Non-exact numerical answers should be given to three significant figures unless stated otherwise.
- You will need a scientific calculator and the Qualitative Analysis Notes provided on pages at the back.

LABORATORY INVESTIGATION OVERVIEW:

This laboratory pack contains authentic Cambridge Advanced Practical examinations. Each practical scenario provides complete procedures, chemical specifications (FA codes), hazard warnings, structured data collection tables, graphical plotting grids, and comprehensive multi-step mathematical and observational deductions.

Question 1: Discrimination and Identification of Aqueous Halide Ions: FA 1, FA 2, and FA 3

Cambridge Syllabus Reference: 9701/33/O/N/23/Q3 · [20 Marks]

Qualitative Analysis Instructions:

You are provided with three aqueous solutions labeled FA 1, FA 2, and FA 3. Each solution contains a sodium salt of a different halide ion: Cl^- , Br^- , or I^- .

Reagents Provided:

- Dilute nitric acid, HNO_3 , 1.0 mol dm^{-3}
- Aqueous silver nitrate, AgNO_3 , 0.05 mol dm^{-3}
- Dilute aqueous ammonia, NH_3 , 1.0 mol dm^{-3}
- Concentrated aqueous ammonia, conc. NH_3 , 8.0 mol dm^{-3} (fume cupboard)
- Chlorine water (or acidified aqueous sodium chlorate(I))
- Cyclohexane solvent (organic solvent for non-polar partition).

Hazard & Safety Advice:

- Silver nitrate stains skin and clothing black. Concentrated ammonia is choking: dispense only in the fume cupboard.
- Cyclohexane is highly flammable. Keep away from naked flames.

(a) Carry out the specified tests on FA 1, FA 2, and FA 3. Record all observations in the table below. [8]

Test	Observations	Deductions
Precipitation Test: To 1 cm depth of solution, add 5 drops of dilute HNO_3 , then add 5 drops of aqueous AgNO_3 .		
Ammonia Solubility Test: To the precipitate from the above test: (i) Add excess dilute $\text{NH}_3(\text{aq})$. (ii) If precipitate remains, add excess conc. $\text{NH}_3(\text{aq})$.		
Halogen Displacement & Organic Partition: To 1 cm depth of solution, add 1 cm depth of chlorine water, then add 1 cm depth of cyclohexane and shake vigorously.		

(b) Identify the halide ion present in each solution based on your experimental observations. [3]

.....

.....

.....

(c) State the purpose of adding dilute nitric acid (HNO_3) before adding aqueous silver nitrate. [2]

.....

.....

.....

(d) Write an ionic equation, with state symbols, for the precipitation of silver bromide from FA 2. [2]

.....

.....

.....

(e) When solid samples of sodium chloride, sodium bromide, and sodium iodide are treated with concentrated sulfuric acid in a fume cupboard, distinct observations are recorded.

State the observations for (i) solid NaBr and (ii) solid NaI, and explain why NaI reduces concentrated H₂SO₄ to a lower oxidation state than NaBr does. [5]

Question 2: Identification of Oxoanions and Gas Evolution: FA 4, FA 5, and FA 6**Cambridge Syllabus Reference: 9701/32/M/J/23/Q3 · [20 Marks]**

Qualitative Analysis Instructions:

You are provided with three solutions: FA 4, FA 5, and FA 6.

Each solution contains one of the following anions: CO_3^{2-} , SO_4^{2-} , or NO_3^- .

Reagents Provided:

- Dilute hydrochloric acid, HCl , 1.0 mol dm^{-3}
- Aqueous barium chloride, BaCl_2 , 0.1 mol dm^{-3}
- Aqueous sodium hydroxide, NaOH , 2.0 mol dm^{-3}
- Aluminium foil pieces
- Fresh limewater, $\text{Ca(OH)}_2(\text{aq})$
- Red and blue litmus papers.

(a) Perform the diagnostic tests specified below and record your observations and deductions. [8]

Test	Observations	Deductions
Acid Reaction: To 1 cm depth of FA 4, add 1 cm depth of dilute HCl . Test any gas with limewater.		
Barium Test: To 1 cm depth of FA 5, add 5 drops of dilute HCl , then add 1 cm depth of aqueous BaCl_2 .		
Nitrate Reduction Test: To 1 cm depth of FA 6, add 1 cm depth of aqueous NaOH , add a small piece of aluminium foil, and warm gently. Test gas with damp red litmus paper.		
Control Comparison: Repeat the Barium Test on FA 4 and FA 6.		

(b) Identify the anion in FA 4, FA 5, and FA 6. [3]

.....

.....

.....

(c) Write an ionic equation, with state symbols, for the reaction between carbonate ions in FA 4 and hydrogen ions from the acid. [2]

.....

.....

.....

(d) Explain why dilute hydrochloric acid must be added before aqueous barium chloride when testing for sulfate ions. [2]

.....

.....

.....

(e) In the nitrate reduction test on FA 6, write the ionic equation for the reduction of nitrate ions by aluminium in alkaline medium. [3]

.....

.....

.....

.....

(f) Describe a chemical test that could distinguish between a solution containing sodium sulfate (Na_2SO_4) and a solution containing sodium sulfite (Na_2SO_3). [2]

.....

.....

.....

Qualitative Analysis Notes

Tables for the reactions of aqueous cations, anions, and tests for gases (Cambridge 9701 Syllabus Reference)

1. Reactions of Aqueous Cations

Cation	Reaction with NaOH(aq)	Reaction with NH ₃ (aq)
Aluminium, Al ³⁺	White ppt., soluble in excess giving a colorless solution	White ppt., insoluble in excess
Ammonium, NH ₄ ⁺	Ammonia gas produced on warming (turns damp red litmus blue)	No reaction
Barium, Ba ²⁺	Faint white ppt. or no ppt. (high [Ba ²⁺])	No ppt.
Calcium, Ca ²⁺	White ppt. with high [Ca ²⁺], insoluble in excess	No ppt. with dilute solutions
Chromium(III), Cr ³⁺	Grey-green ppt., soluble in excess giving dark green solution	Grey-green ppt., insoluble in excess
Copper(II), Cu ²⁺	Pale blue ppt., insoluble in excess	Pale blue ppt., soluble in excess giving intense deep blue solution
Iron(II), Fe ²⁺	Green ppt., insoluble in excess; turns brown on standing near surface	Green ppt., insoluble in excess; turns brown on standing
Iron(III), Fe ³⁺	Red-brown ppt., insoluble in excess	Red-brown ppt., insoluble in excess
Magnesium, Mg ²⁺	White ppt., insoluble in excess	White ppt., insoluble in excess
Manganese(II), Mn ²⁺	Off-white ppt., insoluble in excess; rapidly darkens to brown	Off-white ppt., insoluble in excess; darkens to brown
Zinc, Zn ²⁺	White ppt., soluble in excess giving a colorless solution	White ppt., soluble in excess giving a colorless solution

2. Reactions of Aqueous Anions

Ion	Test	Observation
Carbonate, CO ₃ ²⁻	Add dilute acid	Effervescence; gas turns limewater milky (CO ₂)
Chloride, Cl ⁻ (aq)	Add dilute HNO ₃ , then aqueous AgNO ₃	White ppt., soluble in dilute aqueous ammonia
Bromide, Br ⁻ (aq)	Add dilute HNO ₃ , then aqueous AgNO ₃	Cream ppt., insoluble in dilute NH ₃ , soluble in conc. NH ₃
Iodide, I ⁻ (aq)	Add dilute HNO ₃ , then aqueous AgNO ₃	Yellow ppt., insoluble in dilute and conc. aqueous NH ₃
Nitrate, NO ₃ ⁻ (aq)	Warm with aqueous NaOH, then add Al foil	Ammonia produced; pungent gas turns damp red litmus blue
Sulfate, SO ₄ ²⁻ (aq)	Add dilute HNO ₃ , then aqueous Ba(NO ₃) ₂ / BaCl ₂	Dense white ppt. (insoluble in dilute strong acids)
Sulfite, SO ₃ ²⁻ (aq)	Add dilute acid, warm gently and test gas	Gas produced (SO ₂) turns acidified K ₂ Cr ₂ O ₇ from orange to green

3. Tests for Gases

Gas	Test and Test Result
Ammonia, NH ₃	Turns damp red litmus paper blue; choking pungent smell
Carbon dioxide, CO ₂	Gives a white precipitate with limewater (calcium hydroxide solution turns milky)
Chlorine, Cl ₂	Bleaches damp litmus paper white; pale green choking gas
Hydrogen, H ₂	'Pops' with a lighted wooden splint
Oxygen, O ₂	Relights a glowing wooden splint
Sulfur dioxide, SO ₂	Turns acidified potassium dichromate(VI) paper from orange to green

Examiner Mark Scheme & Indicative Observation Guide

Confidential Marking Points & Error Tolerances for Inorganic Chemistry Practical 3: Qualitative Analysis – Anion Identification & Gas Testing

Question 1: Discrimination and Identification of Aqueous Halide Ions: FA 1, FA 2, and FA 3 [20 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	FA 1 with AgNO ₃ : White precipitate [1]; dissolves completely in dilute aqueous ammonia to give colorless solution [1]; with Cl ₂ + cyclohexane: upper organic layer remains colorless / pale green [1]. FA 2 with AgNO ₃ : Cream precipitate [1]; insoluble in dilute NH ₃ , but dissolves in concentrated NH ₃ [1]; with Cl ₂ + cyclohexane: upper organic layer turns orange / reddish-brown (dissolved Br ₂) [1]. FA 3 with AgNO ₃ : Yellow precipitate [1]; insoluble in both dilute and concentrated NH ₃ [1]; with Cl ₂ + cyclohexane: upper organic layer turns bright violet / purple (dissolved I ₂) [1].	[8]
(b)	FA 1 contains Chloride, Cl ⁻ [1]; FA 2 contains Bromide, Br ⁻ [1]; FA 3 contains Iodide, I ⁻ [1].	[3]
(c)	To react with and remove any carbonate (CO ₃ ²⁻), hydrogencarbonate, or sulfite (SO ₃ ²⁻) impurities [1], which would otherwise form insoluble silver precipitates (e.g. Ag ₂ CO ₃ , Ag ₂ SO ₃) giving a false positive test for halides [1].	[2]
(d)	Ag ⁺ (aq) + Br ⁻ (aq) → AgBr(s) [2] (1 mark for species, 1 mark for state symbols).	[2]
(e)	(i) NaBr + conc. H ₂ SO ₄ : Steamy fumes (HBr) and brown/orange gas/fumes (Br ₂) with choking gas (SO ₂) [1]. (ii) NaI + conc. H ₂ SO ₄ : Purple vapor / dark grey-black solid (I ₂), yellow solid (S), and rotten egg smell (H ₂ S gas) [1]. Explanation: Iodide (I ⁻) has a larger ionic radius and lower charge density than bromide (Br ⁻) [1], so valence electrons are more weakly held / shielding is greater [1], making iodide a significantly stronger reducing agent capable of reducing sulfur from +6 to -2 (H ₂ S), whereas bromide can only reduce sulfur to +4 (SO ₂) [1].	[5]

Question 2: Identification of Oxoanions and Gas Evolution: FA 4, FA 5, and FA 6 [20 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	FA 4 with HCl: Rapid effervescence / bubbling [1]; gas produced turns limewater milky / cloudy white precipitate forms [1]. FA 5 with BaCl ₂ + HCl: Dense white precipitate forms [1], which remains insoluble upon addition of excess dilute HCl [1]. FA 6 with NaOH + Al + heat: Effervescence / gas evolved with pungent choking smell [1]; turns damp red litmus paper blue [1]. Control test: FA 4 with BaCl ₂ gives white ppt that dissolves in HCl; FA 6 gives no precipitate with BaCl ₂ [2].	[8]
(b)	FA 4 contains Carbonate, CO ₃ ²⁻ [1]; FA 5 contains Sulfate, SO ₄ ²⁻ [1]; FA 6 contains Nitrate, NO ₃ ⁻ [1].	[3]
(c)	CO ₃ ²⁻ (aq) + 2H ⁺ (aq) → CO ₂ (g) + H ₂ O(l) [2] (1 mark for balanced equation, 1 mark for state symbols).	[2]
(d)	To eliminate / decompose carbonate (CO ₃ ²⁻) or sulfite (SO ₃ ²⁻) ions which would also react with Ba ²⁺ (aq) to form white precipitates of BaCO ₃ or BaSO ₃ , giving a false-positive result for sulfate [2].	[2]
(e)	Nitrate is reduced to ammonia gas, and aluminium is oxidized to aluminate [1]. Equation: 3NO ₃ ⁻ (aq) + 8Al(s) + 5OH ⁻ (aq) + 18H ₂ O(l) → 3NH ₃ (g) + 8[Al(OH) ₄] ⁻ (aq) [2] (or 8Al + 3NO ₃ ⁻ + 5OH ⁻ + 2H ₂ O → 8AlO ₂ ⁻ + 3NH ₃).	[3]
(f)	Add dilute hydrochloric acid and warm gently: Na ₂ SO ₃ produces sulfur dioxide gas, SO ₂ , which turns acidified potassium dichromate(VI) paper from orange to green [1]; Na ₂ SO ₄ produces no gas and no color change [1].	[2]